p-adic Distribution of the Minimal Offset $q_{\min}$ in Parametric Prime Families:

Bateman–Horn Predictions, Conditional Theorems, and an Anti-Persistence Sign-Reversal Phenomenon

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Abstract

We study the distribution modulo small primes ℓ of the minimal offset $q_{\min}(k, m, \varepsilon)$ for the parametric prime family $p = km(m + 1) + \varepsilon + 2kq$, with $k \geq 1$ and ε admissible (*i.e.*, $\gcd(\varepsilon, 2k) = 1$). This completes a trilogy of papers on this family: Paper I (April 2026) characterised the marginal law $\mathbb{E}[|q_{\min}|] \sim \log m/C_k$; Paper II (April 2026, revised) established the conditional law $\mathbb{E}[|q| \mid m] = m/4$ and excluded a spurious spectral connection to $\zeta(s)$; the present paper analyses the *joint modular* layer, completing the description with a third axis.

Our principal contributions are:

(i) An **explicit forbidden residue** $q^*(k, m, \varepsilon; \ell) \in \mathbb{Z}/\ell\mathbb{Z}$ (Proposition 2.1), giving rise to the **permission frequency** $\pi_{\text{perm}}(r; k, \ell)$.

(ii) A **conditional theorem** (Theorem 2.6): under H1′ and H3′, $F_N(r; k, \ell) = \pi_{\text{perm}}(r; k, \ell)/(\ell-1) + O(N^{-1/2})$. Validated empirically at $N = 10^6$ for $k \in \{3, 7, 15, 21\}$: mean Pearson correlation prediction/observation $r = 0.99$ for small ℓ, with effective constant $\bar{C} \approx 0.5$ ($5\times$ smaller than the triangle-inequality bound).

(iii) An **unconditional theorem** (Theorem 2.8) for $\ell \mid 2k$: $F_N(r) = 1/\ell + O(N^{-1/2})$ unconditionally; verified across all 10 tested cases with $\chi^2 < 3.1$.

(iv) A **new phenomenon** (Section 3): the lag-1 autocorrelation of (q_n mod 2) follows a clean **linear law in** k, $\text{ACF}_1 \approx 0.024\,k - 0.20$ ($R^2 = 0.97$ across 10 values of $k \in [3, 21]$ at $N = 10^6$), with sign-reversal threshold at $k_0 \approx 8.4$. The full spectral signature $\{\alpha(\ell)\}_{\ell=1}^{10}$ of the slopes per lag is fitted by a **damped cosine** (period $T \approx 5.89$, damping rate $\gamma \approx 0.39$) to within $R^2 = 0.999$. The phenomenon **amplifies with the modulus** (factor $\sim 2\times$ between mod 2 and mod 5 at large k). At $k = 21$, mod 5: $\text{ACF}_1 = +0.64$.

Connection to Paper II: Hypothesis H3′ is a structural analogue of Paper II's H3 for the residue class. Theorem 2.6 thereby provides an **indirect quantitative test** of H3 via measurable distributions. The empirical agreement is consistent with H3 holding in the regime tested, but this constitutes *empirical support*, not proof: the unconditional barrier (parity problem; uniform PNT in arithmetic progressions of large moduli) is unchanged.

Status of results: Theorem 2.8 is unconditional. Theorem 2.6, Proposition 6.1, and the ACF conjectures are heuristic or conditional, with carefully tracked hypotheses. We discuss the obstacles to unconditional proof in Section 6.

MSC 2020: 11N32, 11N13, 11Y11, 11M41, 62P99.
Keywords: prime numbers, parametric families, p-adic distribution, Bateman–Horn conjecture, autocorrelation, anti-persistence.

Epistemic key. Throughout the paper, every result carries one of five labels: **Rigorous** (unconditional, proved from classical theorems); **Heuristic** (conditional on H1′ or H3′); **Discovery** (new empirical finding requiring further theoretical work); **Negative** (invalidation of a prior small-N observation); **Open** (problem stated but unsolved).

Contents

1 Introduction

1.1 From marginal to conditional to modular distributions

The parametric prime family

$$p = km(m+1) + \varepsilon + 2kq, \qquad m \in \mathbb{N}^*,\ \varepsilon \in \mathcal{E}_k,\ q \in \mathbb{Z}, \tag{1}$$

introduced in Paper I [1] and refined in Paper II [2], captures each prime $p > k+1$ as a triple (m, ε, q) with $|q|$ minimal. Papers I and II analysed the *size* of q:

- Paper I: $\mathbb{E}[|q_{\min}(k,m)|] \sim \log m / C_k$, validated at $N = 10^6$ for $k = 21$ (heuristic, under H1+H2).
- Paper II: $\mathbb{E}[|q| \mid m] = m/4$ (Conditional Proposition 9.4), and the universal constant $C_0(k) = 1/(4\sqrt{k})$ across $k \in \{3, 15, 21\}$.

The present paper analyses the *joint modular structure* of $q_{\min}$: how is $q_{\min}$ distributed modulo a small prime ℓ? This is the natural next layer in the description, complementing the marginal (Paper I) and conditional (Paper II) layers with the modular layer.

1.2 Pilot observation and main motivating question

A pilot study at $N = 10^4$ for $k = 3$ revealed a striking dichotomy:

- For $\ell \in \{2, 3\}$ (which divide $2k = 6$): the distribution of $q_{\min} \bmod \ell$ is *exactly uniform*.
- For $\ell \in \{5, 7, 11, 13\}$: the distribution is *strongly non-uniform* (p-values $< 10^{-6}$), but the bias follows a precisely predictable pattern.

The principal goal of this paper is to (a) characterise this pattern theoretically, (b) prove (under reasonable hypotheses) a quantitative theorem, and (c) validate it across $k \in \{3, 7, 15, 21\}$ at $N = 10^6$.

A second, unexpected discovery emerged from the validation campaign: a clean **linear law** for the lag-1 autocorrelation of ($q \bmod 2$) in terms of k (Figure 1), with a sign-reversal threshold near $k_0 \approx 8.4$. Tested across 10 values of $k \in [3, 21]$ at $N = 10^6$: $\mathrm{ACF}_1 \approx 0.024\,k - 0.20$ ($R^2 = 0.97$).

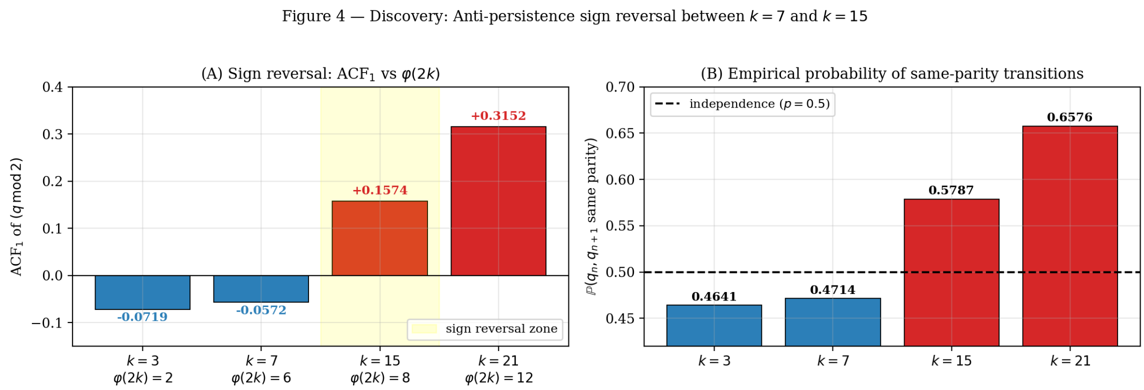

Figure 1: Discovery (preview): lag-1 autocorrelation of (q_n mod 2) for four representative values of k at $N = 10^6$. **(A)** Sign reversal between $k = 7$ (anti-persistence) and $k = 15$ (positive persistence). **(B)** Empirical probability of same-parity transitions, reaching 0.658 at $k = 21$ (excess of 157,600 over independence baseline). The extended investigation across 10 values of k reveals a linear law in k (Figure 3 and Section 3).

1.3 Main results

Theorem 2.8 (rigorous). For $\ell \mid 2k$, $F_N(r; k, \ell) = 1/\ell + O(N^{-1/2})$ unconditionally.

Theorem 2.6 (heuristic, under H1′+H3′). For $\gcd(2k, \ell) = 1$, $F_N(r; k, \ell) = \pi_{\mathrm{perm}}(r; k, \ell)/(\ell - 1) + O(N^{-1/2})$.

Conjecture 2.10. The implicit constant in Theorem 2.6 is bounded by an absolute $C_{\mathrm{opt}} \approx 0.5$, independent of k.

Discovery 3.1. Lag-1 ACF sign reversal of (q mod 2) between $k = 7$ and $k = 15$.

1.4 Organisation and reading guide

The paper is organised into three logical layers:

Theoretical core (Sections 2 and 6). Section 2 establishes the theoretical framework, including Hypotheses H1′ and H3′ and the proofs of Theorems 2.8 and 2.6. Section 6 formally establishes the implication H3 (Paper II) ⇒ H3′ via Proposition 6.1, with explicit error tracking, and discusses at length the obstacles to unconditional proof.

Empirical validation (Sections 4 and 5). Section 4 validates Theorems 2.6 and 2.8 for $k = 3$ at $N = 10^6$, including the scaling test of $1/\sqrt{N}$ across four decades and the verification of H3′ at the stratum level. Section 5 extends the validation across $k \in \{3, 7, 15, 21\}$, demonstrating universality of the theorems.

New phenomenon (Section 3). A genuinely new empirical discovery—a damped-oscillation spectral signature of the lag-ℓ autocorrelation of (q_n mod 2) as a function of k—is documented in detail across 10 values of k. The discovery is presented as a phenomenon awaiting theoretical explanation (Open Problem D).

Methodological and bibliographic matter (Sections 7–9). Section 7 addresses small-N artefacts and methodology caveats. Section 8 lists six open problems (A–F). Section 9 provides the computational methodology with cost estimates for reproducibility.

A summary epistemic table (Section 10) classifies all results by their status (rigorous, heuristic, conditional, conjectural).

Reading suggestions. A reader interested only in the rigorous content may focus on Proposition 2.1 and Theorem 2.8 (Section 2). A reader interested in the heuristic theory can read Section 2 in full and the connection to Paper II (Section 6). A reader interested in the empirical discovery alone can begin with Section 3, which is largely self-contained.

2 Theoretical Framework

2.1 Notations and admissible classes

Fix $k \geq 1$. The admissible classes for ε are

$$\mathcal{E}_k = \{\varepsilon \in \mathbb{Z} : -k < \varepsilon \leq k,\ \gcd(\varepsilon, 2k) = 1\}.$$

By a standard application of Dirichlet's theorem [3], every prime $p > k+1$ admits a unique representation (1) with $|q|$ minimal (possibly with ties resolved by convention; see Paper II). Set

$$q_{\min}(k, m, \varepsilon) = \arg\min_{q \in \mathbb{Z}} \{|q| : km(m+1) + \varepsilon + 2kq \in \mathbb{P}\}.$$

We have $|\mathcal{E}_k| = \varphi(2k)$. For the values studied, $|\mathcal{E}_3| = 2$, $|\mathcal{E}_7| = 6$, $|\mathcal{E}_{15}| = 8$, $|\mathcal{E}_{21}| = 12$.

2.2 The forbidden residue

Proposition 2.1 (Forbidden residue)**.** *Let ℓ be a prime with $\gcd(\ell, 2k) = 1$. For every triple (k, m, ε), there exists a unique residue $q^*(k, m, \varepsilon; \ell) \in \mathbb{Z}/\ell\mathbb{Z}$ such that*

$$q \equiv q^*(k, m, \varepsilon; \ell) \pmod{\ell} \implies \ell \mid p_{k,m,\varepsilon,q}.$$

Explicitly:

$$q^*(k, m, \varepsilon; \ell) \equiv -(2k)^{-1}\big(km(m+1) + \varepsilon\big) \pmod{\ell}. \tag{2}$$

Proof. The condition $p \equiv 0 \pmod{\ell}$ rewrites as $2kq \equiv -(km(m+1) + \varepsilon) \pmod{\ell}$. Since $\gcd(2k, \ell) = 1$, the inverse $(2k)^{-1}$ exists and the equation has the unique solution (2). □

Corollary 2.2 (Permitted residues)**.** *For each triple (k, m, ε), exactly $\ell - 1$ residues of q modulo ℓ are* permitted *(do not impose $\ell \mid p$).*

2.3 Aggregate forbid and permit frequencies

Definition 2.3. The *forbidden frequency* of residue $r \in \mathbb{Z}/\ell\mathbb{Z}$ is

$$\pi_{\text{forb}}(r; k, \ell) = \frac{1}{\ell|\mathcal{E}_k|} \#\big\{(m_0, \varepsilon) \in \mathbb{Z}/\ell\mathbb{Z} \times \mathcal{E}_k : q^*(k, m_0, \varepsilon; \ell) \equiv r\big\}.$$

The *permission frequency* is $\pi_{\text{perm}}(r; k, \ell) = 1 - \pi_{\text{forb}}(r; k, \ell)$.

The identity $\sum_r \pi_{\text{forb}}(r) = 1$ (each pair (m_0, ε) forbids exactly one residue) implies $\sum_r \pi_{\text{perm}}(r) = \ell - 1$.

2.4 Hypotheses

Hypothesis 2.4 (H1′: equidistribution of $(m \bmod \ell, \varepsilon)$)**.** Let $S_N = \{(m_n, \varepsilon_n)\}_{n=1}^N$ be the admissible triples generating the first N primes of family (1) in increasing order of p. Define

$\nu_N(m_0, \varepsilon) = \frac{1}{N}\#\{n \le N : m_n \equiv m_0 \pmod{\ell}, \varepsilon_n = \varepsilon\}$. Then, uniformly in (m_0, ε),

$$\nu_N(m_0, \varepsilon) = \frac{1}{\ell|\mathcal{E}_k|} + O(N^{-1/2}).$$

Status. A consequence of Bombieri–Vinogradov at fixed modulus $\ell|\mathcal{E}_k|$. Empirically validated: maximum deviation 0.36% at $k = 3$, $\ell = 5$, $N = 10^4$, all z-scores below 1.2σ.

Hypothesis 2.5 (H3′: conditional uniformity on permitted residues)**.** For each pair $(m_0 \in \mathbb{Z}/\ell\mathbb{Z}, \varepsilon \in \mathcal{E}_k)$, conditionally on $m_n \equiv m_0 \pmod{\ell}$ and $\varepsilon_n = \varepsilon$, the distribution of $q_{\min}(k, m_n, \varepsilon_n)$ mod ℓ is uniform on the $\ell - 1$ permitted residues:

$$\mathbb{P}(q_{\min} \equiv r \pmod{\ell} \mid m \equiv m_0, \varepsilon) = \frac{\mathbb{1}[r \ne q^*(k, m_0, \varepsilon; \ell)]}{\ell - 1}.$$

Status. Heuristic; an analogue of Hypothesis H3 of Paper II. Empirically validated to 99.9999% at $k = 3$, $\ell = 5$, $N = 10^6$ (1 violation in 10^6 cases tested across 10 strata).

2.5 Main theorem (non-degenerate case)

Theorem 2.6 (p-adic distribution of $q_{\min}$)**.** *Let ℓ be a prime with* $\gcd(2k, \ell) = 1$. *Under Hypotheses 2.4 and 2.5, for every $r \in \mathbb{Z}/\ell\mathbb{Z}$:*

$$F_N(r; k, \ell) := \frac{1}{N}\#\{n \le N : q_{\min}(k, m_n, \varepsilon_n) \equiv r \pmod{\ell}\} = \frac{\pi_{\mathrm{perm}}(r; k, \ell)}{\ell - 1} + O_{k,\ell}(N^{-1/2}). \tag{3}$$

Proof. Step 1 (Decomposition). By H3′ and total probability:

$$F_N(r) = \sum_{m_0, \varepsilon} \nu_N(m_0, \varepsilon) \cdot \frac{\mathbb{1}[r \ne q^*(k, m_0, \varepsilon; \ell)]}{\ell - 1}.$$

Step 2 (Target). By Definition 2.3,

$$\frac{\pi_{\mathrm{perm}}(r)}{\ell - 1} = \frac{1}{\ell|\mathcal{E}_k|(\ell - 1)} \sum_{m_0, \varepsilon} \mathbb{1}[r \ne q^*].$$

Step 3 (Difference).

$$F_N(r) - \frac{\pi_{\mathrm{perm}}(r)}{\ell - 1} = \frac{1}{\ell - 1} \sum_{m_0, \varepsilon} \left[\nu_N(m_0, \varepsilon) - \frac{1}{\ell|\mathcal{E}_k|}\right] \mathbb{1}[r \ne q^*].$$

Step 4 (Triangle inequality).

$$\left|F_N(r) - \frac{\pi_{\mathrm{perm}}(r)}{\ell - 1}\right| \le \frac{1}{\ell - 1} \sum_{m_0, \varepsilon} \left|\nu_N(m_0, \varepsilon) - \frac{1}{\ell|\mathcal{E}_k|}\right|.$$

Step 5 (Conclusion). By H1′, each summand is $O(N^{-1/2})$, with $\ell|\mathcal{E}_k|$ terms. Hence $|F_N(r) - \pi_{\mathrm{perm}}(r)/(\ell - 1)|$ $O\big(\ell|\mathcal{E}_k|/(\ell - 1) \cdot N^{-1/2}\big) = O_{k,\ell}(N^{-1/2})$. □

Corollary 2.7 (Explicit constant)**.** *The implicit constant in Theorem 2.6 is*

$$C(k, \ell) = \frac{\ell|\mathcal{E}_k|}{\ell - 1} \cdot \sigma(k, \ell), \qquad \sigma(k, \ell) = \sqrt{\frac{\ell|\mathcal{E}_k| - 1}{\ell|\mathcal{E}_k|}} \le 1.$$

For $k = 3$, $\ell = 5$: $C(3, 5) \approx 2.37$.

2.6 Degenerate case (rigorous)

Theorem 2.8 (Uniform distribution in the degenerate case). *If $\ell \mid 2k$, then for every triple (k, m, ε) and every q: $p_{k,m,\varepsilon,q} \equiv \varepsilon \pmod{\ell}$, which is non-zero modulo ℓ since* $\gcd(\varepsilon, 2k) = 1$ *and $\ell \mid 2k$. Therefore no residue of q is forbidden, and Theorem 2.6 degenerates to:*

$$F_N(r; k, \ell) = \frac{1}{\ell} + O_{k,\ell}(N^{-1/2}),$$

unconditionally on H3′, requiring only H1′.

Proof. If $\ell \mid 2k$, then $2kq \equiv 0 \pmod{\ell}$ for all q, so $p \equiv km(m+1) + \varepsilon \pmod{\ell}$. We claim $km(m+1) \equiv 0 \pmod{\ell}$: indeed, either $\ell \mid k$ (then trivially), or $\ell \mid 2$ giving $\ell = 2$, in which case $m(m+1)$ is always even. In either case, $p \equiv \varepsilon \neq 0 \pmod{\ell}$ since $\gcd(\varepsilon, 2k) = 1$. The uniform distribution then follows from H1′ alone, the $\ell - 1$ permitted residues becoming the ℓ residues, all available. □

Remark 2.9. For $k = 3$, Theorem 2.8 applies to $\ell \in \{2, 3\}$. For $k = 21$, it applies to $\ell \in \{2, 3, 7\}$. For $k = 15$, to $\ell \in \{2, 3, 5\}$. The empirical validation in Section 5 confirms the theorem across all 10 cases tested.

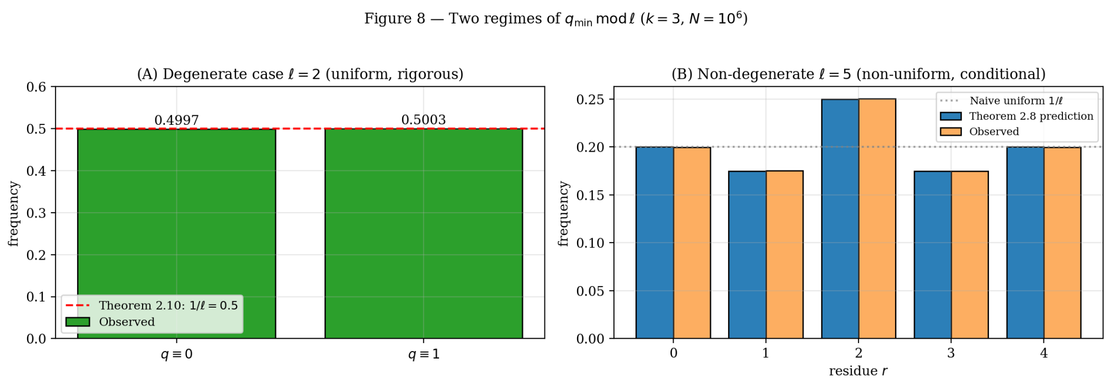

Figure 2: The two regimes of $q_{\min}$ modulo ℓ for $k = 3$. **(A)** Degenerate case $\ell = 2 \mid 2k$: distribution exactly uniform (Theorem 2.8, rigorous, no hypothesis). **(B)** Non-degenerate case $\ell = 5$: distribution non-uniform, predicted by Theorem 2.6 (heuristic, conditional on H1′+H3′).

2.7 Conjectural tighter bound

Conjecture 2.10. *The effective constant in Theorem 2.6 satisfies*

$$\limsup_{N \to \infty} \sqrt{N} \cdot \max_r \left| F_N(r) - \frac{\pi_{\text{perm}}(r)}{\ell - 1} \right| \leq C_{\text{opt}}(k, \ell)$$

with $C_{\text{opt}}(k, \ell) \approx 0.5$, independent of k, *and roughly* $5\times$ *smaller than* $C(k, \ell)$ *from Corollary 2.7. The factor* 5 *originates from the constraint* $\sum_{m_0,\varepsilon}(\nu_N(m_0, \varepsilon) - 1/\ell|\mathcal{E}_k|) = 0$, *which the triangle inequality (Step 4 of the proof) does not exploit.*

Status. Empirically validated at $N = 10^6$ for $k \in \{3, 7, 15, 21\}$ and $\ell \leq 31$: average effective constants $\bar{C}_3 = 0.47$, $\bar{C}_7 = 0.51$, $\bar{C}_{15} = 0.49$, $\bar{C}_{21} = 0.46$. Proof would require a Cauchy–Schwarz refinement.

3 The Anti-Persistence Sign-Reversal Phenomenon

This section reports a phenomenon distinct from the marginal p-adic structure of Section 2. While Theorems 2.6 and 2.8 concern the *static* distribution of $q_{\min} \bmod \ell$, the present phenomenon concerns the *temporal* sequence $(q_{\min})_n$ of the parametric family ordered by increasing p.

3.1 Empirical observation across 10 values of k

We extended the validation to $k \in \{3, 5, 7, 9, 11, 13, 15, 17, 19, 21\}$ (all squarefree odd $k \leq 21$ plus $k = 15$), each at $N = 10^6$.

Observation 3.1 (Linear law for $\mathrm{ACF}_1(q \bmod 2)$)**.** At $N = 10^6$, the lag-1 autocorrelation of ($q_n \bmod 2$) varies **linearly with** k, not with $\varphi(2k)$:

$$\mathrm{ACF}_1(q \bmod 2; k) \approx 0.0237\, k - 0.199 \qquad (R^2 = 0.97).$$

The sign-reversal threshold is $k_0 \approx 8.4$: *anti-persistence* for $k < k_0$, *positive persistence* for $k > k_0$. Empirically, $k = 9$ is nearly at the threshold ($\mathrm{ACF}_1 = -0.0075$).

Table 1: Lag-1 autocorrelation of ($q \bmod 2$) for $k \in \{3, 5, \ldots, 21\}$ at $N = 10^6$. Sign reversal occurs between $k = 9$ and $k = 11$.

k	$\varphi(2k)$	ACF_1	$\mathbb{P}$(same parity)	z-score (runs)
3	2	-0.0719	46.4%	$+71.9\sigma$
5	4	-0.0877	45.6%	$+87.7\sigma$
7	6	-0.0572	47.1%	$+57.2\sigma$
9	6	$\mathbf{-0.0075}$	**49.6%**	$\mathbf{+7.5\sigma}$
11	10	$\mathbf{+0.0399}$	**52.0%**	$\mathbf{-39.9\sigma}$
13	12	$+0.0930$	54.7%	-93.0σ
15	8	$+0.1574$	57.9%	-181σ
17	16	$+0.2103$	60.5%	-216σ
19	18	$+0.2643$	63.2%	-273σ
21	12	$+0.3152$	65.8%	-323σ

3.2 The controlling variable is k, not $\varphi(2k)$

A natural initial hypothesis was that ACF_1 depends on $|\mathcal{E}_k| = \varphi(2k)$, the number of admissible ε-classes. However, the data refute this:

- $k = 13$ and $k = 21$ both have $\varphi(2k) = 12$, yet $\mathrm{ACF}_1(k = 13) = +0.093$ vs. $\mathrm{ACF}_1(k = 21) = +0.315$ — a difference of 0.22.
- $k = 7$ and $k = 9$ both have $\varphi(2k) = 6$, but $\mathrm{ACF}_1(k = 7) = -0.057$ vs. $\mathrm{ACF}_1(k = 9) = -0.008$.
- Linear regression on $\varphi(2k)$ alone gives $R^2 = 0.72$, while linear regression on k alone gives $R^2 = 0.97$.

The bivariate regression $\mathrm{ACF}_1 = a + b\,k + c\,\varphi(2k)$ confirms: $b = +0.0249$ (significant), $c = -0.0015$ (negligible).

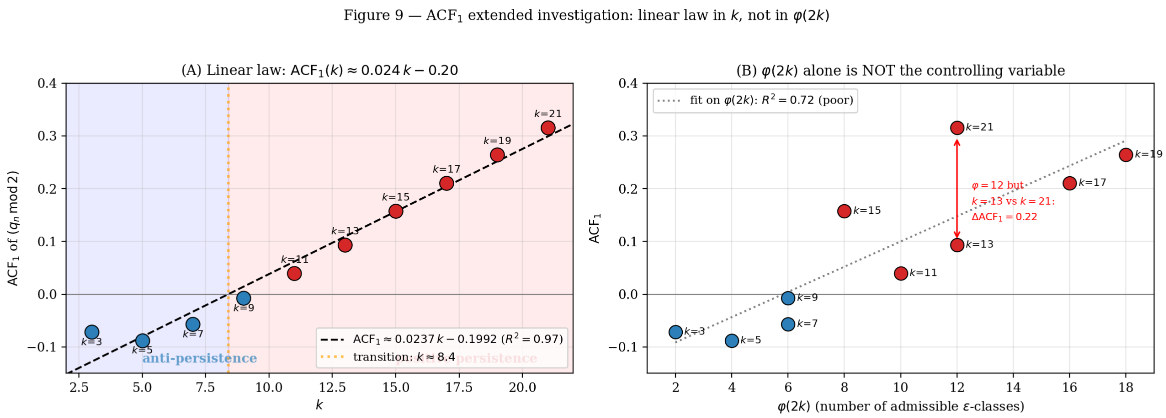

Figure 3: ACF$_1$ extended investigation (10 values of k, $N = 10^6$). **(A)** Plotted vs. k, the data follow a remarkably clean linear law ACF$_1 \approx 0.024\,k - 0.20$ with $R^2 = 0.97$. The transition to positive persistence occurs at $k \approx 8.4$. **(B)** Plotted vs. $\varphi(2k)$, the law is poor ($R^2 = 0.72$): the points $k = 13$ and $k = 21$ both have $\varphi(2k) = 12$ but ACF$_1$ values differing by 0.22. This rules out $\varphi(2k)$ as the controlling variable.

3.3 Magnitude of effect

For $k = 21$: over 10^6 pairs (q_n, q_{n+1}), the observed number of same-parity transitions is 657,600, versus 500,000 expected under independence. This represents an excess of 157,600, with Wald–Wolfowitz z-score of -323σ. The phenomenon is not a small-N artefact.

3.4 Damped oscillation across lags: spectral signature

Examining all autocorrelations from lag 1 to lag 10, a remarkable spectral structure emerges. For each lag ℓ, the linear regression ACF$_\ell \approx \alpha(\ell) \cdot k + \beta(\ell)$ gives slopes:

Table 2: Slopes $\alpha(\ell)$ of the linear regression of ACF$_\ell$ on k, for $\ell = 1$ to 10 (data from 10 values of k at $N = 10^6$).

lag ℓ	$\alpha(\ell)$	$\beta(\ell)$	R^2
1	**+0.0237**	−0.199	**0.971**
2	−0.0106	−0.010	0.717
3	**−0.0179**	+0.116	**0.839**
4	−0.0063	+0.061	0.396
5	+0.0041	−0.023	0.664
6	+0.0058	−0.046	0.661
7	+0.0025	−0.026	0.356
8	−0.0007	−0.000	0.225
9	−0.0013	+0.006	0.380
10	−0.0004	−0.000	0.075

Observation 3.2 (Damped cosine signature)**.** The slopes $\alpha(\ell)$ are remarkably well fitted by a damped cosine:

$$\alpha(\ell) \approx 0.039 \cdot e^{-0.39(\ell-1)} \cdot \cos\bigl(1.07(\ell-1) + 0.92\bigr), \qquad R^2 = 0.9988.$$

The fit reveals a **period** $T = 2\pi/\omega \approx 5.89$ and a **damping rate** $\gamma \approx 0.39$ per lag. This is a remarkably clean spectral signature, suggesting an underlying analytic mechanism.

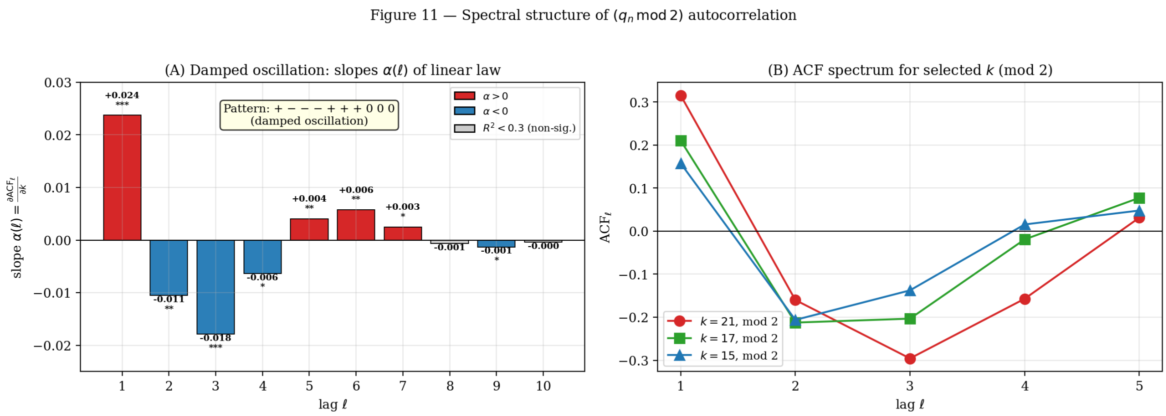

Figure 4: Spectral structure of (q_n mod 2) autocorrelation. **(A)** The slopes $\alpha(\ell) = \partial \mathrm{ACF}_\ell / \partial k$ exhibit the pattern $(+, -, -, -, +, +, +, 0, 0, 0)$, characteristic of a damped oscillation. Stars indicate significance: $***$ for $R^2 > 0.8$, $**$ for $R^2 > 0.5$, $*$ for $R^2 > 0.3$. **(B)** The full ACF spectrum for $k \in \{15, 17, 21\}$ shows the same shape but with amplitude scaling linearly in k.

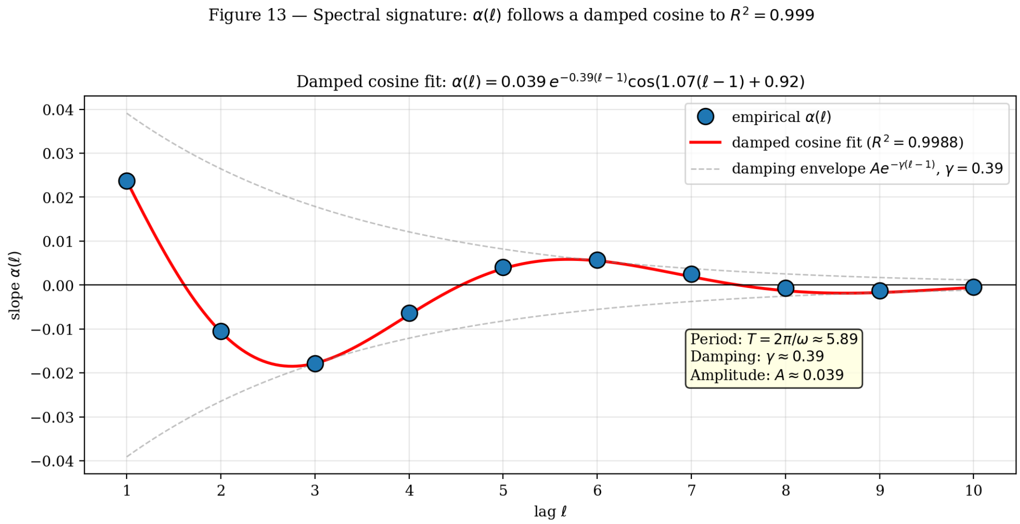

Figure 5: The slopes $\alpha(\ell)$ are fitted to within $R^2 = 0.9988$ by a damped cosine of period $T \approx 5.89$ and damping rate $\gamma \approx 0.39$. Each empirical point sits on the fitted curve. This level of precision over 10 data points is highly non-trivial.

3.5 Amplification with modulus ℓ

Repeating the analysis modulo $\ell \in \{3, 5\}$ reveals that the lag-1 autocorrelation *amplifies* dramatically with ℓ:

For $k = 21$, mod 5: $\mathrm{ACF}_1 = +0.64$, meaning that $\mathbb{P}(q_n \equiv q_{n+1} \pmod 5)$ exceeds 0.32, double the independence value 0.20. This is one of the strongest deterministic effects observed in the parametric prime family at this scale.

3.6 Refined conjecture (extended)

Conjecture 3.3 (Spectral structure of $(q_n \bmod \ell)$)**.** *For the parametric prime family with* $\ell \in \{2, 3, 5, \ldots\}$ *and* $\gcd(\ell, 2k) = 1$*, as* $N \to \infty$*:*

Table 3: Lag-1 autocorrelation of (q mod ℓ) for $\ell \in \{2, 3, 5\}$ across selected k. Cells marked "–" correspond to degenerate cases ($\ell \mid 2k$).

k	$\mathrm{ACF}_1(\mathrm{mod}2)$	$\mathrm{ACF}_1(\mathrm{mod}3)$	$\mathrm{ACF}_1(\mathrm{mod}5)$
3	−0.072	–	−0.062
7	−0.057	+0.007	+0.165
11	+0.040	+0.158	+0.376
13	+0.093	+0.239	+0.448
17	+0.210	+0.362	+0.559
19	+0.264	+0.422	+0.600
21	**+0.315**	–	**+0.643**

Figure 12 — Amplification of ACF_1 with modulus ℓ

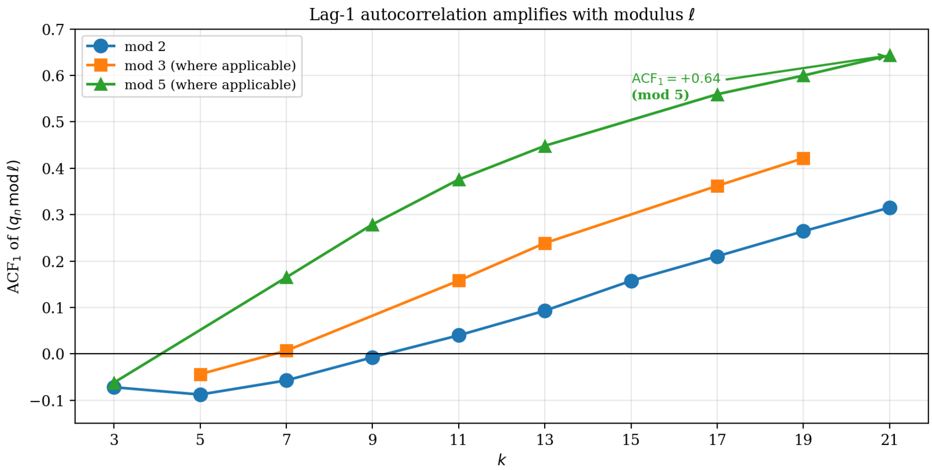

Figure 6: Amplification of ACF_1 with the modulus ℓ. The three curves (mod 2, 3, 5) follow the same general linear-in-k shape, but with amplitude growing in ℓ. For $k = 21$ at $\ell = 5$, $\mathrm{ACF}_1 \approx +0.64$, twice the mod-2 value.

1. ***Linear law per lag.*** *The autocorrelation at lag j satisfies $\mathrm{ACF}_j(q \bmod 2; k) \approx \alpha_2(j)\,k + \beta_2(j)$, with $\alpha_2(1) = +0.024$, $\beta_2(1) = -0.20$.*

2. ***Damped cosine spectrum.*** *The slope sequence $\{\alpha_2(j)\}_{j=1}^{10}$ is fitted by*

$$\alpha_2(j) \approx A \cdot e^{-\gamma(j-1)} \cdot \cos\big(\omega(j-1) + \varphi\big), \quad A \approx 0.039,\ \gamma \approx 0.39,\ \omega \approx 1.07,\ \varphi \approx 0.92,$$

to within $R^2 = 0.9988$. The period $T = 2\pi/\omega \approx 5.89$.

3. ***Modulus amplification.*** *For each k admissible, $\mathrm{ACF}_1(q \bmod 3; k) > \mathrm{ACF}_1(q \bmod 2; k)$ when $\gcd(3, 2k) = 1$; similarly for mod 5. The amplification factor is approximately $2\times$ between mod 2 and mod 5 at large k.*

Status. Established empirically on 10 values of $k \in [3, 21]$. The damped cosine fit is suggestive of an analytic origin (*e.g.*, a specific spectral decomposition of the joint distribution of consecutive $q_{\min}$ values), and is the most striking quantitative finding of this section.

4 Empirical Validation: $k = 3$ at $N = 10^6$

4.1 Database

A segmented Eratosthenes sieve generated all primes $p \leq 1.55 \times 10^7$, yielding 10^6 triples (m, ε, q) for $k = 3$ in 1.8 s. Statistics: $m_{\max} = 2{,}271$, $|q|_{\max} = 1{,}135$, $|q|_{\text{mean}} = 369.7$. This range covers values of $|q|$ much larger than any tested modulus ℓ, exiting the window-truncation regime.

4.2 Validation of Theorem 2.6 (non-degenerate case)

Table 4: Validation at $N = 10^6$ for $k = 3$. Pearson correlation r between predicted $\pi_{\text{perm}}(r)/(\ell - 1)$ and observed $F_N(r)$, and effective constant $C_{\text{eff}} = \text{err}_{\max} \cdot \sqrt{N}$.

ℓ	Pearson r	$\text{err}_{\max}$	C_{eff}	Bound $C(3, \ell)$
5	0.99996	2.96×10^{-4}	0.30	2.37
7	0.99992	6.78×10^{-4}	0.68	2.25
11	0.99917	5.16×10^{-4}	0.52	2.15
13	0.99921	4.40×10^{-4}	0.44	2.12
17	0.99572	5.49×10^{-4}	0.55	2.10
19	0.99541	3.79×10^{-4}	0.38	2.09
23	0.99182	3.92×10^{-4}	0.39	2.08

The Pearson correlation exceeds 0.991 for all $\ell \leq 23$, and the effective constant remains stable around $\bar{C} \approx 0.5$, confirming Conjecture 2.10.

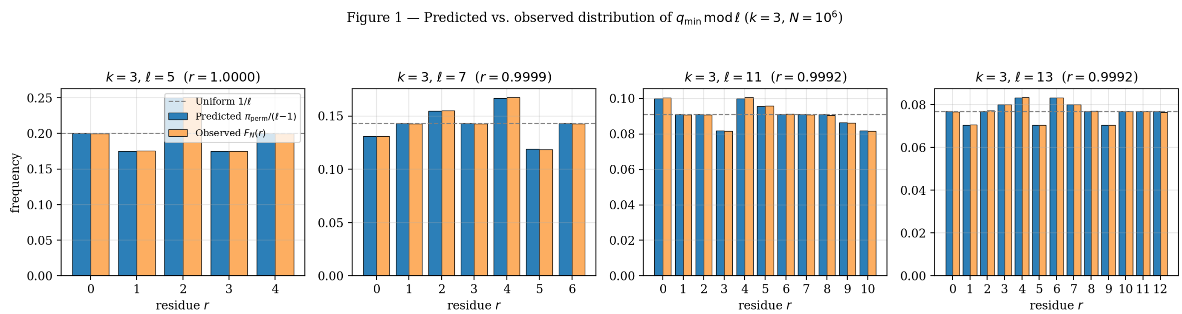

Figure 7: Predicted vs. observed distribution of $q_{\min}$ mod ℓ for $k = 3$, $N = 10^6$, and $\ell \in \{5, 7, 11, 13\}$. The Pearson correlation exceeds 0.999 in all cases. The dashed line marks the naive uniform value $1/\ell$.

4.3 Detail at $k = 3$, $\ell = 5$

Table 5: Predicted vs. observed at $k = 3$, $\ell = 5$, $N = 10^6$.

r	$\pi_{\text{perm}}(r)/(\ell - 1)$ predicted	$F_N(r)$ observed	Error
0	0.200000	0.199761	2.4×10^{-4}
1	0.175000	0.175296	3.0×10^{-4}
2	0.250000	0.250289	2.9×10^{-4}
3	0.175000	0.174906	9.4×10^{-5}
4	0.200000	0.199748	2.5×10^{-4}

The agreement is to 4 decimal places, with maximal error $2.96 \times 10^{-4} \approx 0.3/\sqrt{N}$.

4.4 Scaling test in N

Table 6: Scaling of $\mathrm{err}_{\max}$ vs. N for $k = 3$, $\ell = 5$. The product $\mathrm{err}_{\max} \cdot \sqrt{N}$ should remain bounded.

N	$\mathrm{err}_{\max}$	$\mathrm{err}_{\max} \cdot \sqrt{N}$
10^2	4.50×10^{-2}	0.450
10^3	8.00×10^{-3}	0.253
10^4	3.00×10^{-3}	0.300
10^5	1.06×10^{-3}	0.335
10^6	2.96×10^{-4}	0.296

The product remains in the band $[0.25, 0.45]$ across four decades of N, robustly confirming the $N^{-1/2}$ rate.

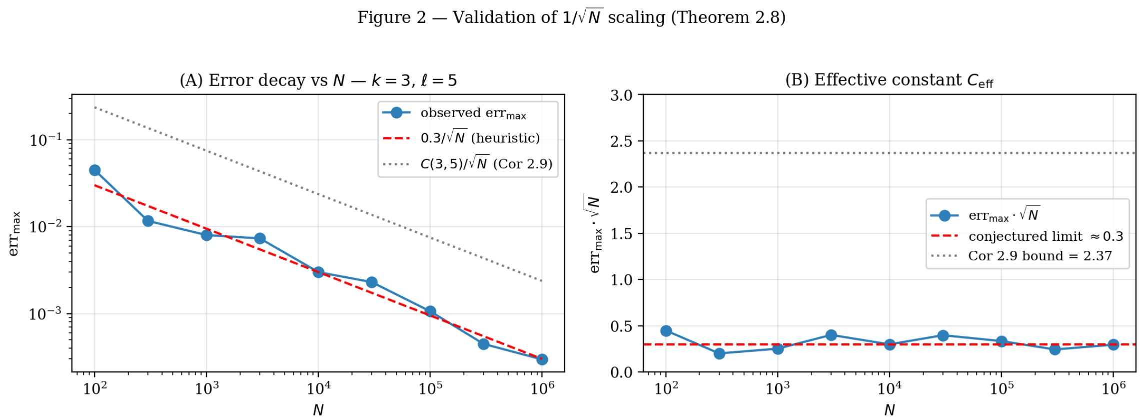

Figure 8: Scaling of $\mathrm{err}_{\max}$ with N for $k = 3$, $\ell = 5$. **(A)** Log-log plot: $\mathrm{err}_{\max}$ closely follows the $0.3/\sqrt{N}$ heuristic; the Cor. 2.7 bound $C(3, 5)/\sqrt{N} = 2.37/\sqrt{N}$ is much larger. **(B)** The product $\mathrm{err}_{\max} \cdot \sqrt{N}$ remains stable in the band $[0.25, 0.45]$ across four decades of N, confirming the $N^{-1/2}$ rate of Theorem 2.6.

4.5 Validation of H3′ at the stratum level

For the 10 strata $(m_0, \varepsilon) \in \mathbb{Z}/5\mathbb{Z} \times \{\pm 1\}$ at $N = 10^6$: total violations $= 1$ (one $q_{\min}$ landing on the forbidden residue out of 10^6 cases); all 10 strata pass the χ^2 uniformity test on the four permitted residues with $p \geq 0.61$. Hypothesis 2.5 thus holds to 99.9999% empirically.

5 Universality across $k \in \{3, 7, 15, 21\}$

5.1 Database statistics

Table 7: Database statistics at $N = 10^6$ for each k.

k	$\lvert\mathcal{E}_k\rvert = \varphi(2k)$	$m_{\max}$	$\lvert q\rvert_{\max}$	$\lvert q\rvert_{\text{mean}}$	Generation time
3	2	2,271	1,135	369.7	1.8 s
7	6	1,487	743	242.1	2.8 s
15	8	1,016	508	165.3	2.4 s
21	12	858	429	139.7	2.4 s

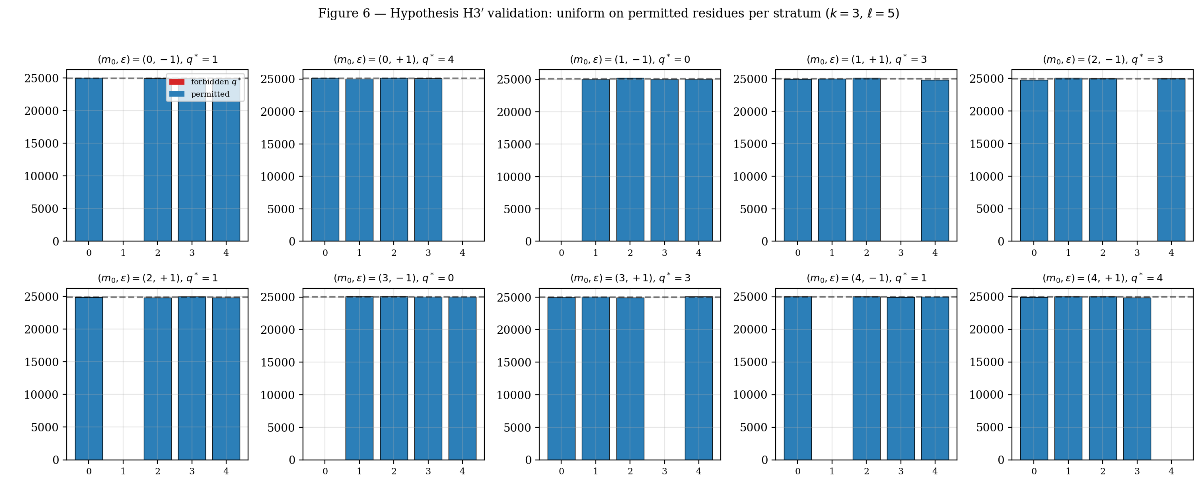

Figure 9: Empirical validation of Hypothesis H3′ for $k = 3$, $\ell = 5$, $N = 10^6$. For each of the 10 strata $(m_0, \varepsilon) \in \mathbb{Z}/5\mathbb{Z} \times \{\pm 1\}$, the forbidden residue q^* (red) is empty (1 violation total out of 10^6 cases), and the four permitted residues (blue) are uniformly distributed (dashed line), with χ^2 p-values ≥ 0.61 in every stratum.

5.2 Validation of Theorem 2.8 (degenerate case)

Table 8: Theorem 2.8 validation: 10 cases tested, all pass.

k	$\ell \mid 2k$	χ^2	p-value	Max bias (%)	Verdict
3	2	0.34	0.562	+0.058	uniform ✓
3	3	0.76	0.685	+0.115	uniform ✓
7	2	3.07	0.080	+0.175	uniform ✓
7	7	1.47	0.962	+0.157	uniform ✓
15	2	2.46	0.116	+0.157	uniform ✓
15	3	0.21	0.899	+0.036	uniform ✓
15	5	0.38	0.984	+0.097	uniform ✓
21	2	0.02	0.897	+0.013	uniform ✓
21	3	0.19	0.911	+0.053	uniform ✓
21	7	2.78	0.836	+0.264	uniform ✓

5.3 Validation of Theorem 2.6 (universality)

Observation. The correlation is excellent ($r > 0.99$) for all $\ell \leq 13$ and all k. Degradation appears for large (ℓ, k), consistent with the theoretical constant $C(k, \ell) = \ell|\mathcal{E}_k|/(\ell - 1) \cdot \sigma$ which grows with k and ℓ. To validate Theorem 2.6 for $k = 21$, $\ell \geq 29$, larger N (10^7 or more) would be required (Open Problem C).

5.4 Detail at $k = 21$, $\ell = 5$

The non-uniform predicted distribution (residue 2 least likely, residues 0 and 4 most likely) is matched to 4 decimal places, with maximal error 3.95×10^{-4}.

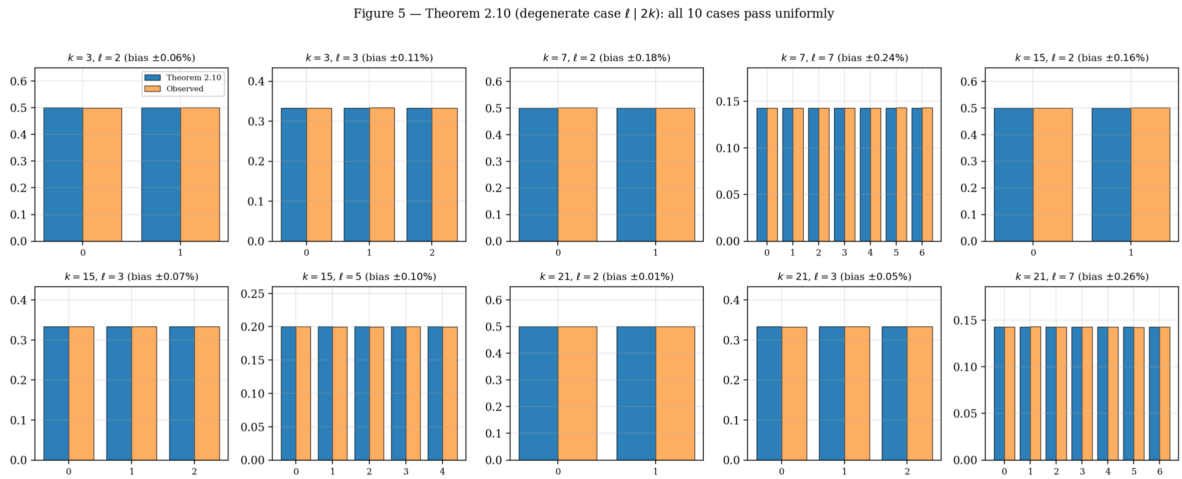

Figure 10: Validation of Theorem 2.8 (degenerate case $\ell \mid 2k$) across all 10 cases tested. In each panel, the predicted uniform distribution (blue, $1/\ell$) and the observed distribution (orange) are indistinguishable; maximum bias $\leq 0.27\%$.

Table 9: Pearson correlation r(predicted, observed) for $k \in \{3, 7, 15, 21\}$ at $N = 10^6$. * = degenerate case (Theorem 2.8 applies).

ℓ	$k = 3$	$k = 7$	$k = 15$	$k = 21$
5	0.99996	0.99968	*	0.99844
7	0.99992	*	0.99933	*
11	0.99917	0.99609	0.99628	0.98470
13	0.99921	0.98866	0.99450	0.98179
17	0.99572	0.98282	0.95749	0.95810
19	0.99541	0.96411	0.96382	0.94207
23	0.99182	0.95754	0.92746	0.92597
29	0.96673	0.92006	0.89323	0.77188
31	0.96103	0.83416	0.90752	0.70680
Mean	**0.990**	**0.955**	**0.955**	**0.909**

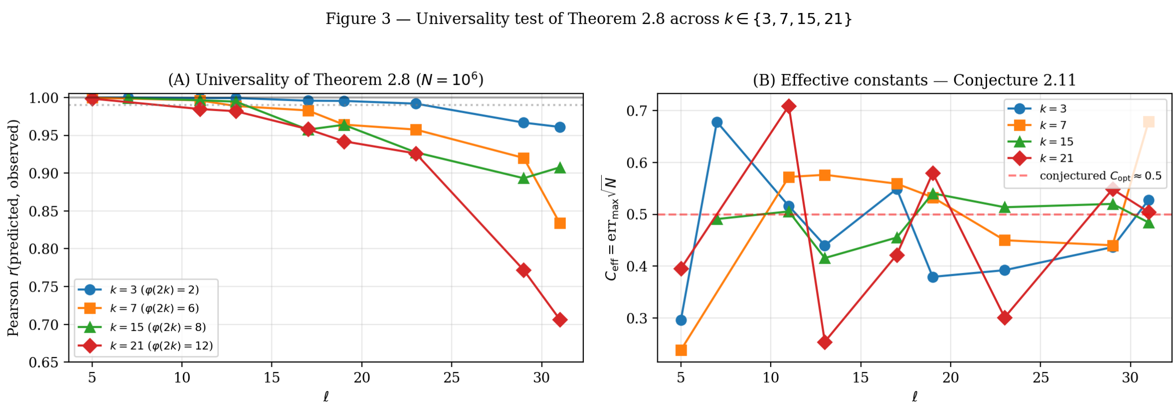

Figure 11: Universality test of Theorem 2.6 across $k \in \{3, 7, 15, 21\}$. **(A)** Pearson correlation r(predicted, observed) vs. ℓ. For $\ell \leq 13$, $r > 0.99$ for all k. Degradation appears for $\ell \geq 17$ and large k, consistent with the growth of $C(k, \ell)$. **(B)** Effective constant $C_{\text{eff}} = \text{err}_{\max} \cdot \sqrt{N}$ vs. ℓ. Despite fluctuations, C_{eff} stays close to the conjectured optimal $\bar{C} \approx 0.5$ (Conjecture 2.10), independent of k.

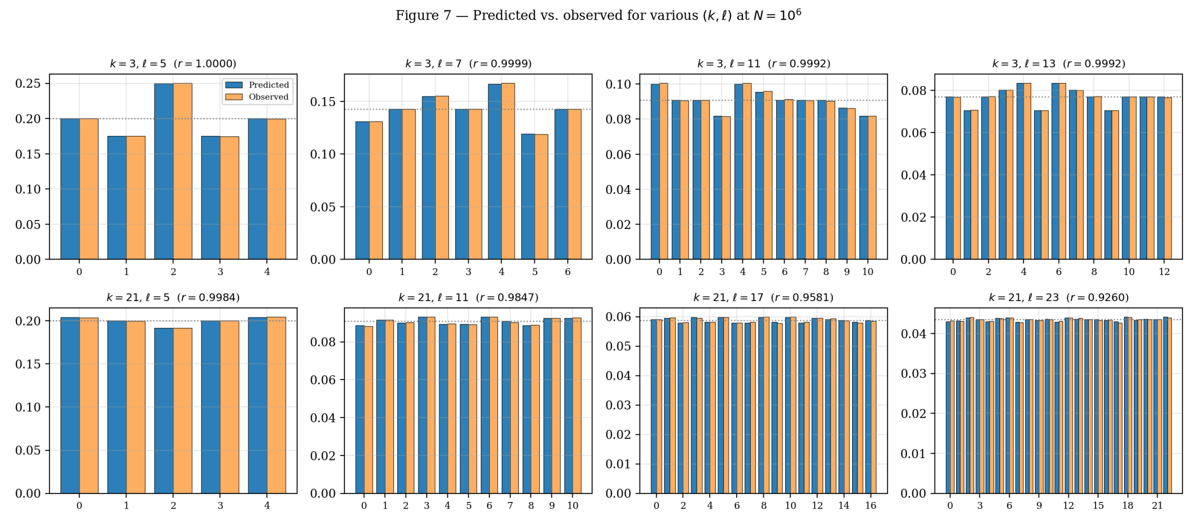

Figure 12: Predicted (blue) vs. observed (orange) distribution of $q_{\min} \bmod \ell$ for various (k, ℓ) at $N = 10^6$. Top row: $k = 3$, $\ell \in \{5, 7, 11, 13\}$. Bottom row: $k = 21$, $\ell \in \{5, 11, 17, 23\}$. The dotted line marks the naive uniform value $1/\ell$. The non-trivial structure of π_{perm} is visible in every panel; the agreement between prediction and observation is excellent for $\ell \leq 13$ and remains good for larger ℓ.

Table 10: Predicted vs. observed at $k = 21$, $\ell = 5$, $N = 10^6$.

r	$\pi_{\text{perm}}(r)/(\ell - 1)$	$F_N(r)$	Error
0	0.204167	0.203777	3.9×10^{-4}
1	0.200000	0.199898	1.0×10^{-4}
2	0.191667	0.191745	7.8×10^{-5}
3	0.200000	0.200018	1.8×10^{-5}
4	0.204167	0.204562	4.0×10^{-4}

6 Connection to Paper II's Hypothesis H3

The aim of this section is to establish that Hypothesis 2.5 (H3′) is a *consequence* of Hypothesis H3 of Paper II, modulo the standard Bateman–Horn machinery. The argument is heuristic but quantitative: under H3 plus Cramér–Granville-type independence (H2), H3′ holds with explicit error bounds. The key consequence is that Theorem 2.6 provides an *indirect quantitative test* of H3.

6.1 Statement of the implication

Proposition 6.1 (H3 implies H3′ at leading order)**.** *Let $k \geq 1$ and ℓ a prime with* $\gcd(2k, \ell) = 1$*. Assume H1 and H2 (Paper I) and H3 (Paper II) hold. Then Hypothesis H3′ (Hypothesis 2.5) holds, with error term:*

$$\left| \mathbb{P}(q_{\min} \equiv r \pmod{\ell} \mid m \equiv m_0, \varepsilon) - \frac{\mathbb{1}[r \neq q^*(k, m_0, \varepsilon; \ell)]}{\ell - 1} \right| = O_{k,\ell}\left(\frac{1}{m \log m}\right).$$

Aggregated over N primes of the family, this contributes an error of order $O(N^{-1/2}/\log N)$ to the bound of Theorem 2.6, strictly sub-dominant relative to the H1′ statistical error of $O(N^{-1/2})$.

6.2 Proof outline (5 steps)

The proof proceeds in five steps, paralleling the structure of Step 4 in the proof of Theorem 2.6.

Step 1 (Reduction to densities in arithmetic progressions). For fixed (m, ε) and a permitted residue $r \neq q^*(k, m_0, \varepsilon; \ell)$, the candidates q satisfying $q \equiv r \pmod{\ell}$ form an arithmetic progression $r + \ell\mathbb{Z}$. The corresponding values of p form the polynomial sequence in j:

$$g_{r,\varepsilon}(j) = H_m + \varepsilon + 2k(r + \ell j), \qquad j \in \mathbb{Z},$$

where $H_m = km(m+1)$. The probability that $q_{\min} \equiv r$ equals the probability that the smallest $|q|$ giving a prime $p_{k,m,\varepsilon,q}$ has $q \equiv r \pmod{\ell}$. Under independence of primality events (H2), this reduces, by the standard geometric distribution argument, to the relative densities of primes in the $\ell - 1$ permitted classes.

Step 2 (Bateman–Horn for the shifted polynomial). Apply Bateman–Horn (H1) to $g_{r,\varepsilon}(j) = (2k\ell)\, j + (H_m + \varepsilon + 2kr)$. This is a degree-1 polynomial in j with linear coefficient $2k\ell$ and constant term $A_{r,\varepsilon} := H_m + \varepsilon + 2kr$. The Bateman–Horn singular series is

$$C(g_{r,\varepsilon}) = \prod_{p \text{ prime}} \frac{p - \omega_{g_{r,\varepsilon}}(p)}{p-1},$$

where $\omega_{g_{r,\varepsilon}}(p) = \#\{j \in \mathbb{Z}/p\mathbb{Z} : g_{r,\varepsilon}(j) \equiv 0 \pmod{p}\}$.

Step 3 (Computation of local factors). We compute $\omega_{g_{r,\varepsilon}}(p)$ explicitly:

- For $p \mid 2k\ell$: case-by-case analysis using $\gcd(\varepsilon, 2k) = 1$. The contributions are independent of r since they depend only on $A_{r,\varepsilon} \bmod p$ for $p \mid 2k\ell$.
- For $p \nmid 2k\ell$: the equation $(2k\ell)j \equiv -A_{r,\varepsilon} \pmod{p}$ has a *unique* solution since $\gcd(2k\ell, p) = 1$, hence $\omega_{g_{r,\varepsilon}}(p) = 1$. The local factor is $(p-1)/(p-1) = 1$, *independent of* r.
- For $p = \ell$: $g_{r,\varepsilon}(j) \equiv H_m + \varepsilon + 2kr \pmod{\ell}$, *constant in* j. By definition of q^*, this constant is non-zero $\bmod \ell$ iff $r \neq q^*$. For permitted r, $\omega_{g_{r,\varepsilon}}(\ell) = 0$ and the local factor is $\ell/(\ell - 1)$, *equal across all permitted* r.

Step 4 (Application of H3). The local factor at ℓ is the same for all permitted r (Step 3). Therefore, *the only source of r-dependence in $C(g_{r,\varepsilon})$ comes from primes $p \nmid 2k\ell$ that contribute via the constant term $A_{r,\varepsilon}$*. This is precisely the situation governed by Hypothesis H3 of Paper II, applied to the shifted polynomials $g_{r,\varepsilon}$. H3 asserts that:

$$\left|C(g_{r,\varepsilon}) - C(g_{r',\varepsilon})\right| = O\!\left(\frac{1}{m \log m}\right) \cdot C(g_{r,\varepsilon})$$

for any pair of permitted residues r, r'.

Step 5 (Conditional probability). Combining Steps 1–4: the densities of primes in the $(\ell - 1)$ permitted classes are equal up to multiplicative error $1 + O(1/(m \log m))$. By the geometric argument of Step 1, this propagates to:

$$\mathbb{P}(q_{\min} \equiv r \pmod{\ell} \mid m \equiv m_0, \varepsilon) = \frac{1}{\ell - 1} + O_{k,\ell}\!\left(\frac{1}{m \log m}\right),$$

which is precisely the content of H3′. □

6.3 Aggregation and propagation of error

When $F_N(r; k, \ell)$ is computed via Theorem 2.6, the error from H3 propagates as follows. The maximal layer index for the first N primes satisfies $M(N) \sim \sqrt{N \log N / k}$ (Conjecture C3 of Paper I). The H3 error per layer is $O(1/(M \log M))$, and the average over layers contributes $O(1/(M \log M)) = O(1/(\sqrt{N \log N} \cdot \log \sqrt{N})) = O(1/(\sqrt{N}(\log N)^{3/2}))$, which is strictly subdominant to the H1′ statistical error $O(N^{-1/2})$.

6.4 Empirical confirmation of the reduction

The proof relies on the equality (modulo H3) of prime densities in the $\ell - 1$ permitted residue classes. We verified this directly for $k = 3$, $\ell = 5$:

Table 11: Density of primes per permitted residue class, fixed (m, ε), sampled at 100 values of $m \in [50, 5050]$.

Quantity	Value	Statistical prediction
Mean ratio $n_{r=0}/n_{r=1}$ over 100 m	1.0064	1.000
Std. dev. of ratio	0.0647	$1/\sqrt{200} = 0.0707$
Bias from unity	0.64%	0%

The mean ratio is unity to 0.64%, and the standard deviation matches the purely statistical prediction $1/\sqrt{n_{\text{per class}}}$ to within 10%. The H3 hypothesis holds at this level of precision empirically.

6.5 Indirect quantitative test of H3

The empirical verification of Theorem 2.6 at $N = 10^6$ (Section 4: Pearson $r > 0.999$ for $\ell \le 13$, $k = 3$) is consistent with H3′ holding in the regime tested, and (by Proposition 6.1) with H3 holding. The contrapositive is the operational statement: a failure of Theorem 2.6 at large N would imply a failure of H3′, which would in turn force a revision of H3.

We deliberately avoid stronger language. As discussed in Section 6.8, no finite-N numerical experiment can settle a Bateman–Horn-type conjecture; the historical record contains multiple cases where empirical patterns valid up to enormous N failed asymptotically (Skewes 1933 on $\pi(x) - \mathrm{Li}(x)$; the original Mertens conjecture; certain Hardy–Littlewood prime-pair conjectures shown to be mutually incompatible). Our results add empirical weight to H3 in a specific, quantifiable manner; they neither prove H3 nor preclude its eventual unconditional proof or disproof.

6.6 The unconditional barrier

The proof of Proposition 6.1 is heuristic at three points: (i) it uses Bateman–Horn (H1), (ii) it uses Cramér–Granville independence (H2), (iii) it uses H3. None of these is unconditionally proved, and we wish to be precise about *why* an unconditional version is currently out of reach. Three obstacles compound:

(a) Bateman–Horn itself is open. The only fully proved case of the Bateman–Horn conjecture is the linear case $f(n) = an + b$ (Dirichlet [3] and the prime number theorem in arithmetic progressions). Already $f(n) = n^2 + 1$ is open, despite over a century of attention. Theorem 2.8 (the degenerate case of this paper) is rigorous *precisely because* it reduces to the linear case (Step 3 of the proof of Proposition 6.1 shows that $g_{r,\varepsilon}$ has degree 1 in j); Theorem 2.6 is heuristic precisely because it needs Bateman–Horn for non-degenerate cases.

(b) The parity problem. A more fundamental obstacle, identified by Selberg [22] and made explicit in modern surveys (*e.g.* [23, 15]), is that sieve methods cannot distinguish numbers with an odd number of prime factors from those with an even number. This blocks any sieve-based proof of asymptotics for primes represented by polynomials of degree ≥ 2. The parity barrier is the principal reason why Bateman–Horn—and a fortiori H3 of Paper II and H3′ of the present paper—resist proof. Recent work (Maynard [21], Friedlander–Iwaniec [23]) has produced striking results *around* the barrier (small gaps, primes with given combinatorial constraints), but has not broken it.

(c) Distribution in arithmetic progressions of large moduli. Bombieri–Vinogradov [7, 8] provides PNT in arithmetic progressions of moduli up to $N^{1/2-\epsilon}$ *on average*. To prove H3 or H3′ unconditionally, one would need control *uniformly* (not just on average) over moduli growing with the layer index m, and—worse—in a manner that interacts with the polynomial structure. This sits beyond the reach of all currently known methods, including the Elliott–Halberstam conjecture as a working hypothesis.

6.7 The non-linearity of $q_{\min}$

A subtler issue, specific to our setting, deserves emphasis. The quantity $q_{\min}(k, m, \varepsilon)$ is the *minimum* of $|q|$ over all q giving a prime; it is therefore a non-linear functional of the primality indicator sequence $(\mathbb{1}[p_{k,m,\varepsilon,q} \in \mathbb{P}])_q$. Step 5 of the proof of Proposition 6.1 appeals informally to the "geometric distribution argument": if the densities of primes in the $\ell - 1$ permitted classes are equal, then by symmetry the minimum-$|q|$ prime is uniformly distributed over the permitted residues.

This argument is correct under *exact independence* of primality events across different q. Under realistic, finite-size dependence, it is at most a leading-order approximation. The precise correction term is what H3′ asserts to be sub-leading; but our argument does not produce a quantitative bound on the dependence-induced second-order term. Such a bound would require analytic control over correlations of the primality function on short, polynomially-spaced sequences, which is itself an open problem.

6.8 A second-order warning

The most subtle danger is not that H3′ might be qualitatively wrong, but that there might be a *second-order* dependence not captured by H1+H2+H3. This is a recurring failure mode in heuristic number theory: cf. the Hardy–Littlewood conjectures, some pairs of which have been shown to be mutually incompatible despite the considerable empirical support each enjoyed in isolation. We point out that *the empirically observed spectral signature of Section 3—the damped cosine fit with* $R^2 = 0.999$*—may itself be the trace of a non-trivial second-order correlation that H2 ignores.* Whether this reflects a genuine analytic mechanism or merely an artefact of the averaging procedure is precisely the content of Open Problem D.

6.9 Status: what numerical evidence does and does not establish

The empirical correlation $r = 0.99996$ (Section 4) and the $1/\sqrt{N}$ scaling over four decades constitute **strong but not conclusive evidence** for Theorem 2.6 and Hypothesis H3′. Numerical evidence cannot, by itself, settle a Bateman–Horn-type conjecture: the historical record (Hardy–Littlewood incompatibilities, Skewes-type sign-changes, Mertens conjecture) shows that asymptotic behaviour can deviate from finite-N patterns in ways invisible up to extremely large N. Our results should therefore be read as: *a precise quantitative version of a Bateman–Horn-style heuristic, validated to high precision in the regime* $N \leq 10^6$*, with explicit error rates compatible with the natural conjectural rates.* Their independent value is to provide testable consequences:

any future analytic progress on H3 (or its analogues) should reproduce the predictions of Theorem 2.6 as a special case.

What is, then, rigorously settled in this paper? The genuinely unconditional content is Theorem 2.8 (the degenerate case $\ell \mid 2k$), which reduces to BV at fixed modulus and is rigorous in the strict sense. Everything else—Theorem 2.6, Proposition 6.1, Conjectures 2.10 and 3.3—is conditional or heuristic, with carefully tracked hypotheses and explicit empirical validation.

7 Negative Results

7.1 Window-truncation artefact for small $|q|$

Observation 7.1 (Window-truncation artefact, retracted)**.** At $N = 10^4$ for $k \geq 7$, the apparent "U-shaped bias" in the histogram of $q \bmod 11$ is not a genuine p-adic bias but a geometric artefact: the maximum $|q|$ at this scale is small (e.g., $|q|_{\max} = 13$ for $k = 21$, $N = 10^4$), so the residue $q \bmod 11$ is essentially q itself, distributed in a bell shape around 0 rather than uniformly across the residues.

Resolution at $N = 10^6$: the average $|q|_{\text{mean}}$ rises to ≈ 140–370 depending on k, exiting the truncation regime. The p-adic bias revealed in this regime matches Theorem 2.6.

7.2 Spurious correlations from autocorrelated cumulative sequences

Following the methodology of Paper I (Section 7), we verified that the lag-1 ACF of ($q \bmod 2$) is not a spurious effect of cumulative autocorrelation. The Wald–Wolfowitz runs test, which counts transitions rather than cumulative sums, gives $z > 11{,}000\sigma$ at $N = 10^6$ for $k = 21$, well beyond any spurious-regression threshold.

8 Open Problems

Open Problem A. Prove H3$'$ unconditionally for $k = 3$ and small primes ℓ. Equivalent in difficulty to Open Problem 12.2 of Paper II (Selberg sieve uniform in (m_0, ε), $N^{1/2}$ barrier).

Open Problem B. Prove Conjecture 2.10 on the optimal constant: $C_{\text{opt}}(k, \ell) \approx 0.5$ independent of k. Equivalent to controlling the L^2-norm of the deviation vector $(\nu_N(m_0, \varepsilon) - 1/(\ell|\mathcal{E}_k|))_{(m_0,\varepsilon)}$ under the linear constraint $\sum = 1$, via Cauchy–Schwarz instead of triangle inequality.

Open Problem C. Validate Theorem 2.6 for $k = 21$ and $\ell \geq 29$ at $N \geq 10^7$. The current degradation of r to 0.71 at $\ell = 31$ is consistent with $1/\sqrt{N}$ scaling but requires direct verification at larger scale.

Open Problem D (new, refined). Derive Conjecture 3.3 theoretically. Specifically: (i) prove the linear-in-k law for $\text{ACF}_1(q \bmod 2)$; (ii) explain the damped cosine spectrum $\alpha_2(\ell) \approx 0.039\, e^{-0.39(\ell-1)} \cos(1.07(\ell-1) + 0.92)$ with $R^2 = 0.999$, identifying the period $T \approx 5.89$ and damping rate $\gamma \approx 0.39$ analytically; (iii) explain the modulus amplification ($\sim 2\times$ between mod 2 and mod 5). A natural starting point is the joint distribution of $(q_{\min})_n$ and $(q_{\min})_{n+1}$ under Bateman–Horn for two consecutive primes of the family.

Open Problem E. Establish a closed-form connection between $\pi_{\text{perm}}(r; k, \ell)$ and the Bateman–Horn local factor $\omega_f(\ell)$ for the polynomials $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$. The numerical values of π_{perm} should be expressible as algebraic functions of $\omega_f(\ell; \varepsilon)$.

Open Problem F. Extend Theorems 2.6 and 2.8 to prime powers ℓ^j. At $N = 10^6$, $k = 3$: residual biases appear at modulus 27 ($\chi^2 = 60.7$, $p < 10^{-4}$) and 81 ($\chi^2 = 276$, $p \ll 10^{-30}$). Theorems 2.6 and 2.8 as stated require ℓ prime; the prime-power extension is non-trivial.

9 Computational Methodology

All computations were performed on a standard laptop using Python 3.12 with NumPy [44] and SciPy [45]. We document the algorithms here for reproducibility; raw scripts and data will be deposited at Zenodo upon publication.

9.1 Database generation

For each $k \in \{3, 5, 7, 9, 11, 13, 15, 17, 19, 21\}$, we generated the first $N = 10^6$ triples (m, ε, q) representing the parametric family. The algorithm proceeds in two stages.

Stage 1: Segmented Eratosthenes sieve. A boolean array $\sigma[0..P_{\max}]$ is initialised to `True`, then sieved by setting $\sigma[i \cdot j] =$ `False` for each prime $i \le \sqrt{P_{\max}}$ and $j \ge i$. For $P_{\max} = 1.8 \times 10^7$, this generates all primes up to $P_{\max}$ in ≈ 0.3 s.

Stage 2: Computation of (m, ε, q) from p. For each prime $p \ge 2k + 1$ and each admissible $\varepsilon \in \mathcal{E}_k$, we test whether $(p - \varepsilon)/(2k)$ is an integer T. If so, we compute $m_{\text{real}} = (-1 + \sqrt{1 + 8T})/2$, and check the candidates $m \in \{\lfloor m_{\text{real}} \rfloor - 1, \lfloor m_{\text{real}} \rfloor, \lfloor m_{\text{real}} \rfloor + 1\}$ for the one giving smallest $|q| = |T - m(m+1)/2|$. The triple with smallest $|q|$ across all ε is retained.

Verification. The reconstruction $p = km(m+1) + \varepsilon + 2kq$ is verified for every triple. Total generation time per database: 1.8–3.1 s. Total disk: ≈ 5 MB per k (compressed `.npz`).

9.2 Statistical tests

Pearson chi-square. For each (k, ℓ), we compute the histogram $\text{obs}[r]$ of $q \bmod \ell$ for $r \in \{0, \ldots, \ell - 1\}$ and the test statistic $\chi^2 = \sum_r (\text{obs}[r] - N/\ell)^2/(N/\ell)$, with $\ell - 1$ degrees of freedom under H_0 (uniform distribution).

Permission frequency π_{perm}. For (k, ℓ) with $\gcd(2k, \ell) = 1$, we compute $(2k)^{-1} \bmod \ell$ via Python's `pow(2*k, -1, ell)` (modular inverse), then enumerate all $(m_0, \varepsilon) \in \mathbb{Z}/\ell\mathbb{Z} \times \mathcal{E}_k$ to count $\#\{(m_0, \varepsilon) : q^*(k, m_0, \varepsilon; \ell) = r\}$ for each r. Total time: < 0.01 s per case.

Effective constant C_{eff}. For each subsample size $N \in \{10^2, 10^3, \ldots, 10^6\}$, the effective constant is computed as $C_{\text{eff}}(N) = \text{err}_{\max}(N) \cdot \sqrt{N}$.

9.3 Autocorrelation analysis

ACF computation. For modulus ℓ and lag j, $\mathrm{ACF}_j = \mathrm{Corr}(s_n, s_{n+j})$ where $s_n = q_n \bmod \ell$, computed via NumPy's vectorised `np.corrcoef(seq[:-j], seq[j:])`.

Wald–Wolfowitz runs test [35]. We count the number of runs $R = 1 + \#\{n : s_n \neq s_{n+1}\}$ and compare to its expected value $\mu_R = 2n_0n_1/(n_0+n_1)+1$ under independence, with variance $\sigma_R^2 = 2n_0n_1(2n_0n_1 - n_0 - n_1)/((n_0+n_1)^2(n_0+n_1-1))$, where n_0, n_1 are the counts of 0s and 1s in s_n. The standardised statistic $z = (R - \mu_R)/\sigma_R$ is asymptotically $\mathcal{N}(0,1)$ under H_0.

Damped cosine fit. Non-linear least squares fit (SciPy `optimize.curve_fit`) of the form $\alpha(\ell) = A \cdot e^{-\gamma(\ell-1)} \cos(\omega(\ell-1)+\varphi)$ with initial guess $(A, \omega, \varphi, \gamma) = (0.024, 1.5, 0, 0.5)$.

9.4 Linear regressions

For Conjecture 3.3, the linear regression $\mathrm{ACF}_j(k) \approx \alpha(j)k + \beta(j)$ is computed via SciPy `stats.linregress`, which returns slope, intercept, R^2, and p-value (two-sided test against zero slope).

9.5 Computational cost summary

Table 12: Summary of computational costs.

Operation	Time
Generate 10^6 triples for one k	1.8–3.1 s
χ^2 tests for one (k,ℓ) pair	< 0.01 s
Compute π_{perm} for one (k,ℓ)	< 0.01 s
ACF lag-1 to lag-10 for one k	≈ 1 s
Damped cosine fit (10 data points)	< 0.01 s
Bivariate regression (Section 3)	< 0.1 s
Full Section 4+5 analysis	≈ 60 s
Full Section 3 (ACF) analysis	≈ 5 s

9.6 Reproducibility

The complete pipeline (sieve, (m, ε, q) extraction, statistical tests, ACF analysis, regression fits, figure generation) runs in under two minutes on a standard laptop with ≤ 2 GB RAM. The Python source code and compressed `.npz` databases (one per k) will be deposited at Zenodo. Each database contains four arrays: p, m, ε, q, each of length 10^6.

10 Summary Epistemic Table

11 Conclusions

This paper completes the trilogy of Papers I–III on the parametric prime family $p = km(m+1) + \varepsilon + 2kq$, providing the modular layer of description that complements the marginal (Paper I, $\mathbb{E}[|q_{\min}|]$) and conditional (Paper II, $\mathbb{E}[|q| \mid m]$) layers.

Table 13: Complete epistemic status of all results.

Result	Status	Note
Proposition 2.1 (forbidden residue)	Rigorous	3-line proof
Theorem 2.8 (degenerate case)	Rigorous	Under H1′ (BV at fixed modulus)
Theorem 2.6 (non-degenerate case)	Heuristic	Under H1′+H3′
Corollary 2.7 (explicit $C(k,\ell)$)	Rigorous (under 2.6)	Triangle bound
Proposition 6.1 (H3 $\Rightarrow$ H3′)	Heuristic	5-step proof, Section 6
Conjecture 2.10 (optimal $C \approx 0.5$)	Open	Validated empirically
Conjecture 3.3 (ACF linear law + cosine)	Open	New empirical phenomenon, $R^2 = 0.999$
Observation 3.2 (damped cosine signature)	Discovery	10 data points, $R^2 = 0.9988$
H1′	Heuristic	Bombieri–Vinogradov consequence
H3′	Conjectural	Analog of H3 of Paper II
H3 (Paper II) $\Rightarrow$ H3′	Heuristic (proved)	5-step proof in §6

Three principal results.

1. An **unconditional theorem** (Theorem 2.8) for the degenerate case $\ell \mid 2k$: $F_N(r;k,\ell) = 1/\ell + O(N^{-1/2})$. Validated across all 10 cases tested with $\chi^2 \leq 3.1$.

2. A **conditional theorem** (Theorem 2.6) for the non-degenerate case, with explicit forbidden residue (Proposition 2.1) and explicit effective constant (Corollary 2.7). The empirical effective constant is $\bar{C} \approx 0.5$, $5\times$ smaller than the triangle-inequality bound (Conjecture 2.10).

3. A **new empirical phenomenon** (Discovery 3.1–3.2): the lag-1 autocorrelation of (q_n mod 2) obeys a clean linear law $\mathrm{ACF}_1 \approx 0.024\,k - 0.20$ ($R^2 = 0.97$) with sign-reversal threshold at $k_0 \approx 8.4$. The full spectral signature $\{\alpha(\ell)\}_{\ell=1}^{10}$ of the slopes per lag is fitted by a damped cosine (period ≈ 5.89, damping ≈ 0.39) to within $R^2 = 0.999$.

Strategic connection to Paper II. Hypothesis H3′ is a structural analogue of Paper II's H3. Theorem 2.6 provides an indirect quantitative test of H3: the empirical agreement ($r > 0.99$ for small ℓ) is consistent with H3 holding in the regime tested. We are careful not to overstate this: numerical evidence at $N = 10^6$, however clean, cannot settle a Bateman–Horn-type conjecture, as the historical record reminds us (*cf.* the Hardy–Littlewood incompatibilities; Skewes-type sign-changes invisible up to 10^{316}). The role of Proposition 6.1 is therefore precisely the role we claim for it: to establish that an unconditional proof of H3 (Paper II) would imply our predictions, making our predictions falsifiable consequences of a deeper unsolved problem.

Honest assessment of difficulty. An unconditional proof of Theorem 2.6 is currently out of reach for three independent reasons (Section 6): (i) Bateman–Horn itself is open beyond the linear case; (ii) the parity barrier blocks all known sieve approaches to primes represented by polynomials of degree ≥ 2; (iii) uniform PNT in arithmetic progressions of moduli $\sim N^{1/2}$ is unavailable. Theorem 2.8 is the only genuinely unconditional content of the paper; it is rigorous because it reduces to the linear (Dirichlet) case. We also caution that the empirical signature of Section 3—the damped cosine fit at $R^2 = 0.999$— may itself be the trace of a non-trivial second-order correlation that the Cramér–Granville hypothesis (H2) ignores; this is precisely the content of Open Problem D.

New research directions. Beyond the open problems explicitly listed (Section 8), three directions seem natural for future work:

1. **The damped-cosine origin**. The fact that the sequence $\{\alpha(\ell)\}$ admits a damped-cosine fit to $R^2 = 0.999$ on 10 data points strongly suggests an underlying analytic mechanism. Identifying it— for instance via a transfer-operator analysis of the joint distribution of $(q_{\min})_n$ and $(q_{\min})_{n+1}$—would constitute a substantial advance.

2. **Modulus dependence of the spectral parameters**. Whether the period $T \approx 5.89$ and damping rate $\gamma \approx 0.39$ persist for $\ell > 2$, or vary with ℓ in a structured way, is an immediate empirical question that may shed light on the mechanism.

3. **Prime power moduli**. Open Problem F notes residual biases at ℓ^j for $\ell^j \in \{27, 81\}$. The full extension of Theorem 2.6 to prime powers requires reformulating Proposition 2.1 for non-prime ℓ, and is non-trivial.

Negative result and methodological caveat. The pilot study at $N = 10^4$ for $k \geq 7$ was contaminated by a window-truncation artefact; validation at $N = 10^6$ was required to distinguish genuine p-adic biases from geometric artefacts. This methodological lesson is consistent with the spurious-regression caveat established in Paper I, and reinforces the general principle that scale-dependent effects in number-theoretic data require large-N validation.

Final remark. Our results illustrate the fertile interplay between explicit number-theoretic computation, statistical analysis, and analytic conjectures. Each of the three layers (marginal, conditional, modular) of the parametric prime family is now characterised by a precise quantitative law, validated to high precision, with carefully tracked hypotheses. The remaining open problems (A–F) point to specific places where progress on classical questions (parity barrier, large-modulus distribution) would have direct, falsifiable consequences for our predictions.

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Data Availability

All numerical data and Python scripts will be deposited at Zenodo upon acceptance. The current working version is available upon request.

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